

2010 年真题参考答案

一、选择题

(1) B. (2) A. (3) C. (4) D. (5) B. (6) D. (7) A. (8) D.

二、填空题

(9) $C_1 e^{2x} + C_2 \cos x + C_3 \sin x$, 其中 C_1, C_2, C_3 为任意常数.

(10) $y = 2x$. (11) $-2^n(n-1)!$. (12) $\sqrt{2}(e^\pi - 1)$. (13) 3cm/s . (14) 3.

三、解答题

(15) 单调增加区间: $(-1, 0)$ 和 $(1, +\infty)$. 单调减少区间: $(-\infty, -1)$ 和 $(0, 1)$.

极大值 $f(0) = \frac{1}{2}\left(1 - \frac{1}{e}\right)$, 极小值 $f(\pm 1) = 0$.

(16) (I) $\int_0^1 |\ln t| [\ln(1+t)]^n dt \leq \int_0^1 t^n |\ln t| dt (n = 1, 2, \dots)$.

(II) $\lim_{n \rightarrow \infty} u_n = 0$.

(17) $\psi(t) = t^3 + \frac{3}{2}t^2 (t > -1)$.

(18) $\arctan\left(\frac{2\pi}{3} + \frac{\sqrt{3}}{4}\right)$.

(19) $\begin{cases} a = -2, \\ b = -\frac{2}{5}, \end{cases} \quad \text{或} \quad \begin{cases} a = -\frac{2}{5}, \\ b = -2. \end{cases}$

(20) $\frac{1}{3} - \frac{\pi}{16}$.

(21) 证明略.

(22) (I) $\lambda = -1, a = -2$.

(II) $k(1, 0, 1)^T + \left(\frac{3}{2}, -\frac{1}{2}, 0\right)^T$ 为 $Ax = b$ 的通解, 其中 k 为任意常数.

(23) $a = -1, Q = \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}, Q^T A Q = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$